Alkesh K. Srivastava

Ph.D. Candidate in Mechanical Engineering, Temple University
Temple Robotics & Artificial Intelligence Lab (TRAIL)
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Education

Last updated: October 5, 2026

Aug 2023 – 2027	*Doctor of Philosophy, Mechanical Engineering* **Temple University** Advisor: Philip Dames
Jan 2021 – Dec 2022	*Master of Engineering, Robotics* **University of Maryland, College Park** Advisor: Michael Otte
Aug 2016 – Nov 2020	*Bachelor of Technology, Electrical Engineering* **Jaipur Engineering College and Research Centre, Rajasthan Technical University**

Research Appointments

Jul 2026 – present	*Visiting Ph.D. Student, GRASP Laboratory* **University of Pennsylvania**	Host: Vijay Kumar
Jan 2026 – present	*Visiting Researcher* **DEVCOM Army Research Laboratory**	Mentor: Carlos P. Nieto
Aug 2023 – present	*Graduate Research Assistant, TRAIL Laboratory* **Temple University**	Advisor: Philip Dames
Mar 2025 – Aug 2025	*ORAU Journeyman Fellow, Research Associateship Program* **DEVCOM Army Research Laboratory**	Mentor: Carlos P. Nieto
Jun 2021 – Aug 2023	*Graduate Researcher / Faculty Assistant, Motion and Teaming Laboratory* **University of Maryland, College Park**	Advisor: Michael Otte

Academic Honors and Awards

2026 **DARS Student Travel Award**, International Symposium on Distributed Autonomous Robotic Systems.

2026 **Graduate Student Travel Award**, Temple University ($600).

2026 **Finalist**, International Conference on Social Robotics + ART Grand Challenge.

2026 **People's Choice Winner**, Graduate Symposium for Research and Creative Works, Temple University.

2026 **First Place**, Late-Stage Research Category Poster Presentation, College of Engineering, Temple University ($500).

2026 **SICE International Young Author Award (SIYA)**, IEEE/SICE International Symposium on System Integration; one of 10 winners selected from 76 submissions.

2026 **IEEE IES Student & Young Professionals Paper Assistance Award**, IEEE Industrial Electronics Society ($1,500).

2025 **Journeyman Fellowship**, DEVCOM U.S. Army Research Laboratory.

2025 **Second Place**, Late-Stage Research Category Poster Presentation, College of Engineering, Temple University ($400).

2024 **Graduate Student Travel Award**, Temple University ($500).

2024 **Vic Schutz Travel Award**, Temple University ($500).

2022 **MRC Conference Travel Award**, Maryland Robotics Center ($1,000).

2021 **MRC "Pathway to the Ph.D." Fellowship**, Maryland Robotics Center.

2020 **Best Paper Award**, 2nd National Conference on Recent Trends & Smart Technologies in Electrical Engineering.

Peer-Reviewed Conference Publications

1. **Alkesh K. Srivastava**, Aamodh Suresh, Carlos Nieto-Granda, and Philip Dames. "When Information is Worth the Risk: Behavioral Valuation for Hazardous Robotic Exploration." *International Symposium of Robotics Research (ISRR)*, 2026.
2. **Alkesh K. Srivastava**, Jared M. Levin, and Philip Dames. "Relay-Based Coordination for Energy-Efficient Multi-Robot Pickup and Delivery." *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2026.
   [DARS Student Travel Award]
3. **Alkesh K. Srivastava**, George P. Kontoudis, Donald Sofge, and Michael Otte. "Bayesian Networks for Path-Based Sensors: Gathering Information and Path Planning in Communication Denied Environments." *World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, 2026.
4. **Alkesh K. Srivastava**, Jared Michael Levin, Alexander Derrico, and Philip Dames. "DELIVER: A System for LLM-Guided Coordinated Multi-Robot Pickup and Delivery using Voronoi-Based Relay Planning." *IEEE/SICE International Symposium on System Integration (SII)*, 2026.
   [SICE International Young Author Award (SIYA; one of 10 winners selected from 76 submissions)
   IEEE IES Student & Young Professionals Paper Assistance Award (SYPA).]
5. **Alkesh K. Srivastava**, Aamodh Suresh, and Carlos Nieto-Granda. "Behaviorally Adaptive Multi-Robot Hazard Localization in Failure-Prone, Communication-Denied Environments." *IEEE International Symposium on Multi-Robot & Multi-Agent Systems (MRS)*, 2025. Poster.
6. **Alkesh K. Srivastava** and Philip Dames. "Speech-Guided Sequential Planning for Autonomous Navigation using Large Language Model Meta AI 3 (Llama3)." *International Conference on Social Robotics + AI*, 2024.
   [Also published in *Social Robotics*, Lecture Notes in Computer Science, vol. 15562, pp. 158–168, 2025.]
7. **Alkesh K. Srivastava**, George P. Kontoudis, Donald Sofge, and Michael Otte. "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions." *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2022.
   [Also published in *Distributed Autonomous Robotic Systems*, Springer Tracts in Advanced Robotics, pp. 286–299, 2024.]
8. **Alkesh K. Srivastava**, Aashish Tanwar, Varun Joshi, Ram Singh, and Gopal Tiwari. "Acoustic Response of Nearby Objects for Visually Impaired." *National Conference on Recent Trends & Smart Technologies in Electrical Engineering*, 2020.
   [Best Paper Award.]

Peer-Reviewed Workshop Papers and Abstracts

1. **Alkesh K. Srivastava**, Jonathan Diller, Vijay Kumar, and Philip Dames. "Spatial-Semantic Uncertainty in VLM-Based Target Search: Balancing Exploration and Identification." *IROS Workshop on Rethinking Uncertainty for Modern Robotics Paradigms*, 2026.
   [Selected for oral presentation (5 of 33 accepted papers) and consideration for the Workshop Best Paper Award.]
2. **Alkesh K. Srivastava**, George P. Kontoudis, Donald Sofge, and Michael Otte. "Path-Based Sensors: Will the Knowledge of Correlation in Random Variables Accelerate Information Gathering?" *ICRA Workshop on Communication Challenges in Multi-Robot Systems*, 2023.

Manuscripts Under Review and in Preparation

1. **Alkesh K. Srivastava**, Carlos Nieto-Granda, and Philip Dames. "Information Gathering Under Fleet Depletion with Resource-Aware Risk Valuation for Hazardous Exploration." Manuscript under review, 2026.
2. **Alkesh K. Srivastava**, Nathan Wolf-Sonkin, and Philip Dames. "Large-Scale LLM-Driven Expressive Swarm Formation with Control Barrier Function Safety Certification." Manuscript under review, 2026.
3. **Alkesh K. Srivastava** and Philip Dames. "A Kinetic Theory of Encounter-Based Information Propagation in Multi-Robot Systems." Manuscript in preparation.
4. **Alkesh K. Srivastava** and Philip Dames. "Adaptive Transport Structure Selection for Multi-Robot Delivery." Manuscript in preparation.
5. **Alkesh K. Srivastava** and Philip Dames. "Reducing Information Staleness in Multi-Robot Tracking via Structured Information Transport." Manuscript in preparation.
6. **Alkesh K. Srivastava**, Aamodh Suresh, Carlos Nieto-Granda, and Philip Dames. "Adaptive Active Target Tracking via Behavioral Entropy with Tunable Uncertainty Sensitivity." Manuscript in preparation.

Teaching

Spring 2024 **Graduate Teaching Assistant**, ENGR 2332 Engineering Dynamics, ***Temple University***. Instructor: Osman Sayginer.

Fall 2023 **Graduate Teaching Assistant**, ENGR 2332 Engineering Dynamics, ***Temple University***. Instructors: Philip Dames and Haijun Liu.

Fall 2022 **Graduate Teaching Assistant**, ENPM 809K Fundamentals for AI and Deep Learning Framework, ***University of Maryland***. Instructor: George Zaki.

Research Mentoring

Malin Aaron Kussi, B.S. Computer Science, Temple University 2025 – present

Jared Michael Levin, B.S. Electrical and Computer Engineering, Temple University 2024 – 2025
Coauthor on SII 2026 and DARS 2026; Placement: Control Systems Engineer, ***Brock Solutions***

Alexander Derrico, B.S. Mechanical Engineering, Temple University 2024 – 2025
Coauthor on SII 2026; Placement: Engineering Drafter & Project Manager, ***Monarch Precast Concrete***

Professional Service

Journal and Magazine Reviewer

IEEE Robotics and Automation Letters (RA-L), 5 reviews, 2025–present.

IEEE Consumer Electronics Magazine (CEM), 2 reviews, 2026–present.

Conference Reviewer

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 3 reviews, 2026–present.

IEEE International Conference on Robotics and Automation (ICRA), 3 reviews 2024–present.

IEEE International Symposium on Multi-Robot & Multi-Agent Systems (MRS), 2025–present.

World Symposium on the Algorithmic Foundations of Robotics (WAFR), 2026–present.

Professional Memberships

IEEE Member.

IEEE Robotics and Automation Society (RAS) Member.

IEEE Industrial Electronics Society (IES) Member.

IEEE Young Professionals Member.

Society of Instrument and Control Engineers (SICE) Member.

IEEE Robotics and Automation Society Technical Committee on Robot Control Member.

Talks and Presentations

Nov 2026 "When Information is Worth the Risk: Behavioral Valuation for Hazardous Robotic Exploration." Scheduled presentation, *International Symposium of Robotics Research (ISRR)*, Malacca, Malaysia.

Oct 2026 "Relay-Based Coordination for Energy-Efficient Multi-Robot Pickup and Delivery." Scheduled presentation, *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, Tokyo, Japan.

Sep 2026 "Spatial-Semantic Uncertainty in VLM-Based Target Search: Balancing Exploration and Identification." Oral and poster presentation, *IROS Workshop on Rethinking Uncertainty for Modern Robotics Paradigms*, Pittsburgh, PA, USA.

Apr 2026 "Coordination and Collaboration in Multi-Robot Teams: Planning, Creativity, and Interactive Human-Robot Decision-Making." Poster, *Temple University Graduate Symposium for Research and Creative Work*, Philadelphia, PA, USA.

Mar 2026 "Coordination and Collaboration for Resource Transport in Constrained Multi-Robot Systems." Student Seminar, *Temple University*, Philadelphia, PA, USA.

Feb 2025 "Speech-Guided Sequential Planning for Autonomous Navigation using Large Language Model Meta AI 3." Poster, *Temple University*, Philadelphia, PA, USA.

2024 "Speech-Guided Sequential Planning for Autonomous Navigation using Large Language Model Meta AI 3." Poster, *International Conference on Social Robotics (ICSR)+AI*, Odense, Denmark.

Nov 2023 "Estimating Hazard Locations in Communication-Denied Environments: Distributed Multi-Robot Approaches using Path-Based Sensors." Student Seminar, *Temple University*, Philadelphia, PA, USA.

Jun 2023 "Path-Based Sensors: Will Knowledge of Correlation in Random Variables Accelerate Information Gathering?" Oral presentation, *ICRA Workshop on Communication Challenges in Multi-Robot Systems: Perception, Coordination, and Learning*, London, UK.

May 2023 "Estimating Hazardous Locations in Communication-Denied Environments: A Bayesian Network Approach with Path-Based Sensors." Poster, *Maryland Robotics Center Research Symposium*, College Park, MD, USA.

May 2023 "Estimating Hazardous Locations in Communication-Denied Environments: Distributed Multi-Robot Approaches with Path-Based Sensors." Poster, *Maryland Robotics Center Research Symposium*, College Park, MD, USA.

Nov 2022 "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions." Oral presentation, *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, Montbéliard, France.

Nov 2022 "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions." Student Seminar, *Maryland Robotics Center*, College Park, MD, USA.

Feb 2022 "Path Planning for Maximizing Information Gathered When Hazards Correlate with Targets." Student Seminar, *Maryland Robotics Student Seminar*, College Park, MD, USA.